



Bacteria with SONAR designed

Scientists at Caltech have engineered bacterial cells with SONAR to be dispatched within the human body for medical purposes. <http://digit.in/tinysonar>



Plants have wearables now

Researchers have created wearable tattoos sensors for plants to make precise measurements on agricultural water use. <http://digit.in/plantattoo>

AI in healthcare

AI's advent in healthcare is not limited to huge and expensive projects, quite a lot of it is happening very close to you

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rtificial intelligence is the latest in a long list of tech must-have buzzwords – that might have been

overused, according to quite a few actual AI experts. But the level of intelligence aside, our current understanding, and implementations, of AI are capable of bringing about vast improvements in any field it is applied to. That also applies to taking care of yourself, either by yourself or through your local medical practitioners. A wide array of AI-based applications and services are now available that make you quite capable of dealing with certain minor and recurring healthcare needs on your own. Here's a few that caught our attention.

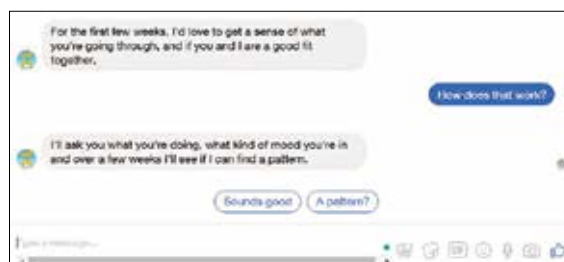
NIRAMAI (BREAST CANCER)

Breast cancer is a leading cause of cancer deaths in women. Early detection is critical to most cases, and an Indian startup, NIRAMAI (Non-Invasive Risk Assessment with Machine Intelligence), has developed a software that can do just that – with much higher accuracy, and much earlier than what traditional methods or self screening are capable of. It's a low-cost, portable screening tool that works without radiation, touch or invasion. NIRAMAI puts AI to work in the backend to deliver accuracy in analysis. The backend software, SMILE, also provides capabilities like automated breast region segmentation that reduces the need for

expertise from the user's end. NIRAMAI has three solutions designed for speciality hospitals, small scale practitioners and large scale screening.

CICER FROM TEN3T (VITAL STATS)

Another medical device startup from India, ten3t has a 9cm patch named Cicer that can fit into the palms of any adult individual that can conduct a 6 lead ECG and track pulse, respiration, blood oxygen, temperature, BP, all the while streaming data in real time. AI algorithms running on the physician's end look for signs in this real time data that indicate risks and act accordingly. Cicer can be effectively used for in-home patient monitoring.



Woebot is a AI based chatbot that analyses your mental health through conversations

WOEBOT, WYSA (DEPRESSION)

After our recent extensive features on everything to do with chatbots, we don't need to tell you all over again how capable they are. When it comes to healthcare, mental health especially benefits a lot from chatbots. There are chatbots being implemented to deal with specific mental health issues and teach you how to achieve a better state of mind yourself. For instance, Woebot, developed by Stanford research-

ers, is just another bot on Facebook Messenger to begin with. Rather than asking about your past, the bot checks in on your mood regularly and tries to notice a pattern. Then, based on cognitive behavioural therapy, it provides you suggestions or even recommends going to a human therapist.

Closer to our shores, Wysa is a similar chatbot developed by a team of Indian entrepreneurs that asks you a series of questions about your current mood, checks in with you regularly, and provides an overall constant companionship that may not be available if you were relying solely on a human therapist. Such chatbots may not be of any use for serious mental ailments, but might actually be just what most people need in initial stages.

AI ASSISTED FITNESS: VI

Perhaps the most futuristic of all, Vi is a smart assistant that lives in a pair of wireless headphones and is targeted to assist your workouts like walking, running, cycling, with meditation, open cardio and HIIT on the way this year. Retailing for \$199, the headphones contain sensors that can detect a bunch of stats during your run like heart rate, cadence and more.

All of this data sharing, as well as the coaching, happens in real time. So, if you're pushing yourself too hard, Vi will tell you to take it down a notch. If you're in your ideal heart rate zone, Vi will help you maintain your form, all the while open to questions and responses from you. The headphones are a pair of Beautiful sound units by Harmon Kardon that offer 6-8 hours of usage and work with both Android and iOS devices. [A](#)